

THEORY AND METHODS

ESTIMATION OF GENERALIZED VARIANCE, PRECISION AND COVARIANCE MATRICES USING THE PITMAN NEARNESS CRITERION

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Summary: The problem of preferring one estimator of Σ over another is considered within the class of estimators $a_i S$, $i = 1, 2$, where $S \sim W_p(n-1, \Sigma)$. Computer simulations are performed to demonstrate the conditions on the a_i , $i = 1, 2$ under the loss function $L(\Sigma, \hat{\Sigma}) = \text{tr } \hat{\Sigma} \Sigma^{-1} + \text{tr } \hat{\Sigma}^{-1} \Sigma - 2p$. The asymptotic probability of closeness of such estimators is obtained. Estimation of the precision matrix and the generalized variance is also considered using the Pitman nearness criterion. Finally an estimator of the signal to noise ratio is studied.

1. INTRODUCTION

Let T_1 and T_2 be two estimators for a parameter θ . Pitman (1937) defined a criterion for which T_1 is preferable to T_2 in estimating θ if

$$(1.1) \quad \text{Prob}\{|T_1 - \theta| < |T_2 - \theta|\} \geq \frac{1}{2}.$$

A multivariate generalization of (1.1) was given by Peddada (1985) as follows. Prefer T_1 over T_2 if

$$(1.2) \quad \text{Prob}\{L(T_1, \theta) < L(T_2, \theta)\} \geq \frac{1}{2}$$

where $L(T, \theta)$ is the loss function associated with the estimation of the $p \times p$ matrix θ by T . The criterion of closeness as defined by Pitman is not transitive, as noted by Blyth (1972). The intransitivity means that in the estimation of θ , if T_1 is preferable to T_2 , and T_2 is preferable to T_3 , then T_1 may not be preferable to T_3 .

In this paper we study the estimators of generalized variance, precision and covariance matrices using the Pitman nearness criterion. In section 2 we study the estimators of Σ and Σ^{-1} using the generalized Pitman nearness criterion. In section 3 we compare estimators of $|\Sigma|$. Section 4 gives asymptotically closer estimators of Σ . Finally in section 5 the estimation of the signal to noise ratio is studied.

2. ESTIMATORS OF Σ AND Σ^{-1}

DEFINITION 2.1. Let θ be a $p \times p$ non-singular matrix of functions of parameters and T be a $p \times p$ corresponding estimator of θ . The estimator T_1 is closer to θ than T_2 in "generalized Pitman" sense under the loss function L , if

$$(2.1) \quad \text{Prob}\{L(T_1, \theta) < L(T_2, \theta)\} \geq \frac{1}{2}.$$

We shall denote this preference of T_1 over T_2 by

$$(2.2) \quad T_1 \gg_p T_2.$$

In the case that the probability of closeness tends to $1/2$ for large n , we use the following:

DEFINITION 2.2. An estimator T_1 for θ is called asymptotically closer than an estimator T_2 under the loss function L if

$$\lim_{n \rightarrow \infty} \sqrt{n} [\text{Prob}\{L(T_1, \theta) < L(T_2, \theta)\} - \frac{1}{2}] > 0.$$

For estimating the covariance matrix Σ , we shall take hereafter the loss function (see Gupta and Krishnamoorthy, 1990; Krishnamoorthy and Gupta, 1989; Gupta and Ofori-Nyarko, 1994)

$$(2.3) \quad L(\hat{\Sigma}, \Sigma) = \text{tr } \hat{\Sigma} \Sigma^{-1} + \text{tr } \hat{\Sigma}^{-1} \Sigma - 2p$$

where $\hat{\Sigma}^{-1}$ is well-defined with probability one for $n > p + 1$, and restrict attention to specific classes

$$(2.4) \quad C_1 \equiv \{cS: 0 < c < 1, S \sim W_p(n-1, \Sigma)\}$$

$$(2.5) \quad C_2 \equiv \{cS^{-1}: c > 1, 2, S \sim W_p(n-1, \Sigma)\}$$

LEMMA 2.1. Let $a_i S \in C_i$, $i = 1, 2$; be two estimators of Σ such that $0 < a_2 < a_1 < 1$. Then, $a_1 S >>_p a_2 S$ provided $\text{med}(u_1) > a_1 a_2$ where $u_1 = \text{tr}(S^{-1}\Sigma)/\text{tr}(S\Sigma^{-1})$.

Proof: Under the loss function (2.3) and in the class C_1 , $a_1 S >>_p a_2 S$ iff

$$(2.6) \quad \text{Prob}\left\{a_1 \text{tr } S\Sigma^{-1} + \frac{1}{a_1} \text{tr } S^{-1}\Sigma < a_2 \text{tr } S\Sigma^{-1} + \frac{1}{a_2} \text{tr } S^{-1}\Sigma\right\} \geq \frac{1}{2}.$$

After simplifying (2.6), we obtain

$$(2.7) \quad \text{Prob}\left\{a_1 a_2 < \frac{\text{tr } S^{-1}\Sigma}{\text{tr } S\Sigma^{-1}}\right\} \geq \frac{1}{2}.$$

Notice that $\text{tr } S^{-1}\Sigma = \text{tr } \Sigma^{\frac{1}{2}} S^{-1} \Sigma^{\frac{1}{2}}$ and is distributed as the trace of an inverted Wishart $W_p^{-1}(n-p, I)$. Also $\text{tr } S\Sigma^{-1} = \text{tr } \Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}}$ which is distributed as $\chi_{(n-1)p}^2$. Let $u_1 = \text{tr } S^{-1}\Sigma/\text{tr } S\Sigma^{-1}$, then (2.7) holds provided

$$(2.8) \quad \text{med}(u_1) > a_1 a_2$$

where $\text{med}(u_1)$ is the median of the random variable u_1 . ■

LEMMA 2.2. Let $a_1 > a_2 > 1$, and $a_i S^{-1} \in C_2$ ($i = 1, 2$) be two estimators of Σ^{-1} , then $a_1 S^{-1} >>_p a_2 S^{-1}$ if and only if

$$(2.9) \quad \text{med}(u_1) \leq \frac{1}{a_1 a_2}.$$

Proof: (2.9) can be proved similarly as outlined in the proof of Lemma 2.1. ■

We have indicated that the estimator $a_1 S$ is preferable, in the GPN sense, to $a_2 S$ for which $0 < a_2 < a_1 < 1$ provided $\text{med}(u_1) \geq a_1 a_2$. The next theorem shows the estimators $a_i S$ are themselves improved by unique scalar multiples of S . The following definition is helpful in proving the theorem (see Khattree, 1987).

DEFINITION 2.3. Within the class C_1 , let $aS \in C_1$. We say aS is locally superior in the GPN sense, if for arbitrary small Δa in the neighbourhood of a ,

$$(2.10) \quad aS >>_p (a + \Delta a)S.$$

By taking the limit as $\Delta a \rightarrow 0$, locally superior estimators can be found.

THEOREM 2.1. Under the loss function (2.3) we have

(i) taking the limit as $\Delta a \rightarrow 0^+$, aS is locally superior to its right neighbours iff

$$(2.11) \quad a^2 \geq \text{med}(u_1).$$

(ii) taking the limit as $\Delta a \rightarrow 0^-$, aS is locally superior to its left neighbours iff

$$(2.12) \quad a^2 \leq \text{med}(u_1).$$

Consequently, (2.11) and (2.12) lead to a unique choice of $a_0 S$ which is locally superior to left as well as right neighbours.

Proof: We will prove (i) only. Part (ii) can be proved similarly. Note that, $aS \gg_p (a + \Delta a)S$, iff

$$(2.13) \quad \text{Prob} \left\{ a \text{tr} S \Sigma^{-1} + \frac{1}{a} \text{tr} S^{-1} \Sigma \leq (a + \Delta a) \text{tr} S \Sigma^{-1} + \frac{1}{(a + \Delta a)} \text{tr} S^{-1} \Sigma \right\} \geq \frac{1}{2}.$$

After simplifying, we find that for $\Delta a > 0$, (2.13) becomes

$$(2.14) \quad \text{Prob} \left\{ \frac{\text{tr} S^{-1} \Sigma}{\text{tr} S \Sigma^{-1}} \leq a(a + \Delta a) \right\} \geq \frac{1}{2}.$$

Taking the limit as $\Delta a \rightarrow 0^+$, $\text{Prob} \left\{ \frac{\text{tr} S^{-1} \Sigma}{\text{tr} S \Sigma^{-1}} \leq a^2 \right\} \geq \frac{1}{2}$ iff $\text{med}(u_1) \leq a^2$.

By proving (ii) we obtain the condition $a^2 \leq \text{med}(u_1)$. Thus, a unique choice of a is

$$(2.15) \quad a_0 = \sqrt{\text{med}(u_1)}$$

which gives the improved estimator $a_0 S$. ■

THEOREM 2.2. For estimating Σ^{-1} in the class C_2 , the optimum choice a_0 under the loss function (2.3) is given by

$$(2.16) \quad a_0 = \frac{1}{\sqrt{\text{med}(u_1)}}.$$

The estimators obtained from (2.15) and (2.16) have desirable properties. In the class C_1 , any estimator aS except $a_0 S$ can be improved upon.

One can choose any number c in (a, a_0) or (a_0, a) such that cS is an improvement over aS in the GPN sense. Within C_1 , no improvement is possible in a_0S . Thus we have the following theorem.

THEOREM 2.3. The estimator a_0S is the best unique estimator of Σ within the class C_1 under the loss function (2.3) in the sense that for any other estimator aS within C_1 , a_0S cannot be improved within C_1 .

Proof: Suppose $\exists a_1$ in C_1 for which $a_1S \gg_p a_0S$, $0 < a_0 < a_1 < 1$. Using (2.8) and (2.15), we see that $a_0(a_0 - a_1) \geq 0$. It follows that $a_0 > a_1$, which is a contradiction. ■

If an estimator outside C_1 needs to be compared with those within C_1 , it is necessary to compare it with a_0S . Table 1 gives the median of the random variable u_1 , for selected values of $n = 10, 20, 50$ and $p = 2, 3, 5$. The median of u_1 was evaluated using a computer program by Smith and Hocking (1972) where 10,000 Wishart random variables were generated.

TABLE 1. Median of $u_1 = \frac{\text{tr } S^{-1}\Sigma}{\text{tr } S\Sigma^{-1}}$

n	p		
	2	3	5
10	0,01369	0,01526	0,01532
20	0,00291	0,00304	0,00339
50	0,00043	0,00043	0,00042

3. COMPARISON OF ESTIMATORS OF $|\Sigma|$

Here we consider the class $C_3 \equiv \{c|S|, 0 < c < 1, S \sim W_p(n-1, \Sigma)\}$. Results similar to the previous section are obtained.

LEMMA 3.1. Let $a_i |S|$, $i = 1, 2$ be two estimators of $|\Sigma|$ such that $0 < a_2 < a_1 < 1$, then under the loss

$$(3.1) \quad L(|\hat{\Sigma}|, |\Sigma|) = |\Sigma^{-1}\hat{\Sigma}| + |\hat{\Sigma}^{-1}\Sigma| - 2,$$

$a_1 |S| \gg_p a_2 |S|$ iff $\text{med}(v) \leq \sqrt{\frac{1}{a_1 a_2}}$, where $v = |S|/|\Sigma|$.

Proof: Note that $\text{Prob}\{L(a_1 |S|, |\Sigma|) < L(a_2 |S|, |\Sigma|)\} \geq \frac{1}{2}$, iff

$$\text{Prob}\left\{|S|/|\Sigma| \leq \frac{1}{\sqrt{a_1 a_2}}\right\} \geq \frac{1}{2}.$$

■

Here $|S|/|\Sigma| = |\Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}}|$ and $\Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}} \sim W_p(n-1, I)$ and $v = |\Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}}|$ is distributed as the product of p independent chi-square random variables with $(n-i)$ degrees of freedom $i = 1, 2, \dots, p$. The density function of v is not easy to compute. Anderson (1984) has given the density of v for special cases when $p = 1$ or $p = 2$ (see also Gupta and Tang, 1986, 1988).

Table 2 gives the median of the random variable $v = |\Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}}|$ (for $p = 2, 3, 5$ and $n = 20, 50, 100$). The median of v was evaluated using a computer program by Smith and Hocking (1972) where 10,000 Wishart random variables were generated.

TABLE 2. Median of $v = |\Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}}|$

n	p		
	2	3	5
20	342,83	297,61	193,42
50	2348,23	2231,88	1919,67
100	9658,85	9459,89	8821,35

4. ASYMPTOTICALLY CLOSER ESTIMATORS

Let $p \times 1$ vectors $X_1, X_2, \dots, X_n$ be a random sample from a p -variate normal distribution $N_p(0, \Sigma)$ and put $S = \sum_{i=1}^n X_i X_i'$. Some estimators of the covariance matrix Σ are compared using the concept of asymptotic probability of closeness.

DEFINITION 4.1. An estimator T_1 for Σ is called asymptotically closer than an estimator T_2 under the loss function L if

$$(4.1) \quad \lim_{n \rightarrow \infty} \sqrt{n} [\text{Prob}\{L(T_1, \Sigma) < L(T_2, \Sigma)\} - \frac{1}{2}] > 0.$$

We now state the main result of this section.

THEOREM 4.1. Let S have a Wishart distribution $W_p(n, \Sigma)$. Then the estimator $\frac{S}{\sqrt{n(n-p-1)}}$ is asymptotically uniformly closer than $\frac{S}{n+p+1}$ under the loss (2.3). More precisely we obtain

$$(4.2) \quad \text{Prob}\left[L\left[\frac{S}{\sqrt{n(n-p-1)}}, \Sigma\right] < L\left[\frac{S}{n+p+1}, \Sigma\right]\right] = \frac{1}{2} + \frac{9p(p+1)-4}{24\sqrt{\pi p n}} + o(n^{-1}).$$

Proof: Let $D \equiv L\left[\frac{S}{\sqrt{n(n-p-1)}}, \Sigma\right] < L\left[\frac{S}{n+p+1}, \Sigma\right]$. Note that D can

be expressed in terms of $Y = \sqrt{n} \left[\frac{S}{n} - \Sigma \right]$ as

$$(4.3) \quad D = \frac{\left[\left[1 + \frac{p+1}{n} \right] - \left[1 - \frac{p+1}{n} \right] \right]^{\frac{1}{2}}}{(n+p+1) \left[1 - \frac{p+1}{n} \right]^{\frac{1}{2}}} \operatorname{tr} [\sqrt{n} Y \Sigma^{-1} + nI] \\ + n \left[\left[1 - \frac{p+1}{n} \right]^{\frac{1}{2}} - \left[1 + \frac{p+1}{n} \right] \right] \operatorname{tr} [\sqrt{n} Y \Sigma^{-1} + nI]^{-1}.$$

Now we have

$$(4.4) \quad \operatorname{tr} [\sqrt{n} Y + nI]^{-1} \\ = \frac{1}{n} \left[p - \frac{1}{\sqrt{n}} \operatorname{tr} Y + \frac{1}{n} \operatorname{tr} Y^2 - \frac{1}{n\sqrt{n}} \operatorname{tr} Y^3 + \frac{1}{n^2} \operatorname{tr} Y^4 + O_p(n^{-\frac{5}{2}}) \right].$$

Using (4.4), and taking $\sqrt{n(n-p-1)} \doteq n - \frac{1}{2}(p+1)$, D can be simplified as

$$D = \frac{3(p+1)}{n\sqrt{n}} \operatorname{tr}(Y \Sigma^{-1}) - \frac{3}{2} \frac{(p+1)}{n^2} \operatorname{tr}(Y \Sigma^{-1})^2 - \frac{3}{4} \frac{p(p+1)^2}{n^2} + O_p(n^{-\frac{5}{2}}).$$

Hence the characteristic function of $\frac{n\sqrt{n} D}{3(p+1)}$ can be written as

$$(4.5) \quad E \left\{ \exp \left[\frac{it n\sqrt{n} D}{3(p+1)} \right] \right\} \\ = E e^{it \operatorname{tr}(Y \Sigma^{-1})} \left\{ 1 + \frac{it}{\sqrt{n}} \left[-\frac{\operatorname{tr}(Y \Sigma^{-1})^2}{2} - \frac{p(p+1)}{4} \right] \right\} + O(n^{-1}).$$

Each expectation in (4.5) can be evaluated by using the results of Sugiura and Fujimoto (1982), giving

$$\begin{aligned}
 (4.6) \quad E\left\{\exp\left[\frac{it}{3(p+1)}n\sqrt{n}D\right]\right\} &= e^{-t^2p}\left[1 - \frac{it}{\sqrt{n}}\frac{p(p+1)}{4}\right]\left[1 + \frac{4p}{3\sqrt{n}}(it)^3\right] \\
 &\quad - \frac{it}{2\sqrt{n}}(p(p+1) + 4(it)^2p)e^{-t^2p} + o(n^{-1}) \\
 &= e^{-t^2p}\left[1 - \frac{3}{4}\frac{p(p+1)}{\sqrt{n}}(it) - \frac{2}{3}\frac{(it)^3}{\sqrt{n}}p\right] + o(n^{-1}).
 \end{aligned}$$

Inversion of the characteristic function yields

$$\begin{aligned}
 (4.7) \quad \text{Prob}\left[\frac{n\sqrt{n}D}{3(p+1)\sqrt{2p}} < x\right] \\
 = \Phi(x) + \frac{3}{4}\frac{(p+1)p}{\sqrt{2pn}}\phi(x) + \frac{1}{3\sqrt{2pn}}(x^2-1)\phi(x) + o(n^{-1})
 \end{aligned}$$

where $\Phi(x)$ and $\phi(x)$ stand for the cumulative distribution and density function of the standard normal distribution respectively. ■

Note that for $x = 0$, we have

$$(4.8) \quad \sqrt{n}\left[\text{Prob}\left\{L\left[\frac{S}{\sqrt{n(n-p-1)}}, \Sigma\right] < L\left[\frac{S}{n+p+1}, \Sigma\right]\right\} - \frac{1}{2}\right] > 0 \text{ and } n \rightarrow \infty.$$

THEOREM 4.2. Under the loss (2.3) the estimator $\frac{S}{\sqrt{n(n-p-1)}}$ is asymptotically closer than $\frac{S}{n}$.

Proof: We can write $D \equiv L\left[\frac{S}{\sqrt{n(n-p-1)}}, \Sigma\right] - L\left[\frac{S}{n}, \Sigma\right]$ as

$$(4.9) \quad D = \frac{1}{4} \frac{p(p+1)^2}{n^2} + \frac{(p+1)}{n\sqrt{n}} \text{tr}(Y\Sigma^{-1}) - \frac{(p+1)\text{tr}(Y\Sigma^{-1})^2}{2n^2} + o_p(n^{-5/2}).$$

The characteristic function of $\frac{n\sqrt{n}D}{p+1}$ is

$$(4.10) \quad E\left[e^{\frac{it n\sqrt{n}D}{p+1}}\right] = Ee^{it\text{tr}(Y\Sigma^{-1})} \left\{ 1 + \frac{it}{\sqrt{n}} \left[-\frac{\text{tr}(Y\Sigma^{-1})^2}{2} + \frac{p(p+1)}{4} \right] \right\} + o(n^{-1})$$

$$= e^{-t^2 p} \left\{ 1 - \frac{p(p+1)}{4\sqrt{n}} it - \frac{2p}{3\sqrt{n}} (it)^3 \right\} + o(n^{-1}).$$

Therefore

$$E\left[e^{\frac{it n\sqrt{n}D}{(p+1)\sqrt{2p}}}\right] = e^{-\frac{t^2}{2}} \left\{ 1 - \frac{p(p+1)}{4\sqrt{2pn}} it - \frac{(it)^3}{3\sqrt{2np}} \right\} + o(n^{-1}).$$

Inversion of the characteristic function gives

$$(4.11) \quad \text{Prob}\left[\frac{n\sqrt{n}D}{(p+1)\sqrt{2p}} < 0\right] = \frac{1}{2} + \frac{3p(p+1) - 4}{24\sqrt{np\pi}} + o(n^{-1}). \quad \blacksquare$$

REMARK. It follows that asymptotically the estimator $\{n(n-p-1)\}^{-\frac{1}{2}}S$ is preferable to $\frac{S}{n}$, which is the m.l.e. of Σ , and to $\frac{S}{n-1}$ which is the unbiased estimator of Σ .

5. ESTIMATION OF SIGNAL TO NOISE RATIO

The estimation of the signal to noise ratio has been considered by Khatri and Rao (1987) by shrinking the Mahalanobis distance. The analysis was carried out assuming that an independent sample from the multivariate normal was given. This section is motivated by the work of Khattree and Gill (1987)

who proposed alternative estimators to that given by Khatri and Rao by using the Pitman nearness criterion. We shall confine ourselves within the class of estimators which are of the form: a positive constant times the Mahalanobis distance.

5.1 Minimum risk criterion

Let δ be a $p \times 1$ multivariate signal and $f^{-1}S$ be the estimated noise covariance matrix of order $p \times p$. Assume the components of δ and S are real. The following theorem is stated in Khatri and Rao (1987).

THEOREM 5.1. *Let δ be a $p \times 1$ vector and $S \sim W_p(f; \Sigma)$; then*

$$\frac{\delta' \Sigma^{-1} \delta}{\delta' S^{-1} \delta} \sim \chi^2(f-p+1), \text{ and } \frac{(\delta' S^{-1} \delta)^2}{(\delta' S^{-1} \Sigma S^{-1} \delta)(\delta' \Sigma^{-1} \delta)} \sim \beta_1\left[\frac{f-p+2}{2}, \frac{p-1}{2}\right],$$

and are independently distributed, where $\beta_1(a, b)$ denotes the beta distribution of the first kind with parameters a and b .

The following result, which we will need, is well known. Let $X \sim \beta_1(a, b)$ and $Y \sim \chi_c^2$. Also let $\mu_r'(X)$ and $\mu_r'(Y)$ be the r -th raw moments of X and Y respectively. Then $EX^r = \frac{\Gamma(a+r)\Gamma(a+b)}{\Gamma(a+r+b)\Gamma(a)}$, and $EY^r = 2^r \Gamma\left[\frac{c}{2} + r\right] / \Gamma\left[\frac{c}{2}\right]$. The noise ratio $\rho(S, \Sigma)$ is defined by

$$\rho(S, \Sigma) = (\delta' S^{-1} \delta)^2 / (\delta' S^{-1} \Sigma S^{-1} \delta).$$

Define the class $C = \{cD_p^2 : c > 0\}$, where $D_p^2 = f\delta' S^{-1} \delta$ is the squared Mahalanobis distance.

THEOREM 5.2. *Consider the loss function $L(\hat{\Sigma}, \Sigma) = \text{tr}(\hat{\Sigma}\Sigma^{-1}) + \text{tr}(\hat{\Sigma}^{-1}\Sigma) - 2p$. Denote the optimum multiplying factor c which minimizes*

the risk $R(\rho(S, \Sigma), cD_p^2)$ within the class C by c . Then the optimum c is given by

$$(5.1) \quad c_* = \frac{1}{f} \left[\frac{(f-p+2)(f-p+1)(f-p-1)(f-p)}{(f+1)(f-1)} \right]^{\frac{1}{2}}$$

and the corresponding risk is

$$(5.2) \quad 2 \left\{ \left[\frac{(f-p+2)(f-p+1)(f-1)}{(f-p)(f-p-1)} \right]^{\frac{1}{2}} - p \right\}.$$

Proof: The problem is to minimize $E\{L(\rho(S, \Sigma), cD_p^2)\}$ with respect to $c > 0$. That is, to minimize

$$(5.3) \quad E \left[\frac{cf\delta'S^{-1}\delta}{\rho(S, \Sigma)} + \frac{\rho(S, \Sigma)}{cf\delta'S^{-1}\delta} \right] - 2p.$$

Differentiating (5.3) with respect to c , and equating to zero and solving,

leads to $c_*^{-2} = E \left\{ \frac{(f\delta'S^{-1}\delta)}{\rho(S, \Sigma)} \right\} / E \left\{ \frac{\rho(S, \Sigma)}{f\delta'S^{-1}\delta} \right\}$. Using the fact that $\rho(S, \Sigma)$

and $\delta'S^{-1}\delta$ are independent, we have

$$(5.4) \quad c_* = \frac{1}{f} \left\{ \frac{E(\rho(S, \Sigma)) E \left[\frac{1}{\delta'S^{-1}\delta} \right]}{E \left[\frac{1}{\rho(S, \Sigma)} \right] E(\delta'S^{-1}\delta)} \right\}^{\frac{1}{2}}$$

$$= \frac{1}{f} \left\{ \frac{(\delta'S^{-1}\delta) \frac{(f-p+2)}{f+1} \cdot \frac{f-p+1}{\delta'S^{-1}\delta}}{\frac{f-1}{f-p} \cdot \frac{1}{\delta'S^{-1}\delta} \cdot \frac{\delta'S^{-1}\delta}{f-p-1}} \right\}^{\frac{1}{2}}$$

$$= \frac{1}{f} \left\{ \frac{(f-p+2)(f-p+1)(f-p-1)(f-p)}{(f+1)(f-1)} \right\}^{\frac{1}{2}}.$$

The corresponding risk is given by

$$\begin{aligned} R(\rho(S, \Sigma), c_* D_p^2) &= c_* f E \left\{ \frac{\delta' S^{-1} \delta}{\rho(S, \Sigma)} \right\} + \frac{1}{c_* f} E \left\{ \frac{\rho(S, \Sigma)}{\delta' S^{-1} \delta} \right\} - 2p \\ &= 2 \left\{ \left[\frac{(f-p+2)(f-p+1)(f-1)}{(f-p)(f-p-1)} \right]^{\frac{1}{2}} - p \right\}. \end{aligned}$$

5.2 Comparison of various estimators

In this section we compare our estimator $c_* D_p^2$ to the estimators obtained by Khattree and Gill (1987) using various loss functions. The optimum c_i , $i = 0, 1, 2, \dots, 4$ using the loss functions

$$\begin{aligned} (5.5) \quad L_0(t, \theta) &= (t - \theta)^2, \quad L_1(t, \theta) = \frac{t}{\theta} - \log \left[\frac{t}{\theta} \right] - 1 \\ L_2(t, \theta) &= \frac{\theta}{t} - \log \left[\frac{\theta}{t} \right] - 1, \quad L_3(t, \theta) = \left[\frac{t}{\theta} - 1 \right]^2, \text{ and} \\ L_4(t, \theta) &= \left[\frac{\theta}{t} - 1 \right]^2 \end{aligned}$$

where t is an estimator of the parameter θ and

$$\begin{aligned} (5.6) \quad c_0 &= (f-p+2)(f-p-3)/f(f+1), \quad c_1 = (f-p-1)(f-p)/f(f-1) \\ c_2 &= (f-p+1)(f-p+2)/f(f+1), \quad c_3 = (f-p-3)(f-p-2)/f(f-3), \text{ and} \\ c_4 &= (f-p+3)(f-p+4)/f(f+3). \end{aligned}$$

Comparison of the estimators $c_i D_p^2$, $i = 0, 1, 2, \dots, 4$ is achieved as follows: One would prefer $b_1 D_p^2$ over $b_2 D_p^2$ if

$$(5.7) \quad \text{Prob}\{ |b_1 D_p^2 - \rho(S, \Sigma)| \leq |b_2 D_p^2 - \rho(S, \Sigma)| \} \geq \frac{1}{2}, \quad (b_1 > b_2)$$

according to the Pitman nearness criterion (1.1). Simplifying (5.7) we obtain

the condition $\frac{f(b_1+b_2)}{4} \leq \text{med}(Y)$, where the distribution of Y involves the confluent hypergeometric function of the second kind (Khatri and Rao, 1987)). The median of Y was evaluated for a given pair (f, p) as in table 3 of Khatri, Rao and Sun (1986). Pairwise comparison of $c_i D_p^2$ and $c_j D_p^2$, is given in table 3, which indicates the proportion of times the row estimator is better than the column estimator. From table 3 it is seen that c_* is better than c_0, c_1, c_2, c_3, c_4 in terms of the relative frequency, followed by c_1 and c_0 .

TABLE 3

Proportion of times the row estimator is better than the column estimator

c_*	0,6818	0,6035	0,064	0,5976	0,0409
c_0		0	0,0691	0,6603	0,0451
c_1			0,0666	0,6257	0,0541
c_2				0,6121	0,0419
c_3					0,0545

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